

Tyler Schmidt

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Research Interests: Bayesian semiparametrics, reinforcement learning, causal inference, simulation-based inference, Bayesian machine learning

Education

Ph.D. in Statistics, University of Iowa

Fall 2025 – Present

Advisor: Prof. Nathan Wikle

- Admitted via accelerated undergraduate-to-graduate (U2G) pathway, Fall 2025; transitioned from M.S. to Ph.D. status, Spring 2026

B.S. in Statistics and B.A. in Mathematics, University of Iowa

Awarded December 2025

- Cumulative GPA: 3.53
- Relevant coursework: Measure-Theoretic Probability, Real Analysis, Theoretical Statistics I-II, Differential Equations, Numerical Analysis.

Research Experience

Research with Prof. Nathan Wikle, Dept. of Statistics

Jan 2025 – Present

University of Iowa

- Established a Bernstein–von Mises calibration result for causal effects defined by stochastic interventions, including incremental propensity score interventions, building on the semiparametric posterior correction framework of Yiu et al. (2025)
- Derived entropy bounds and posterior contraction rates verifying the framework’s sufficient conditions for a soft Bayesian additive regression trees (SoftBART) prior
- Designed and ran a simulation study showing the resulting method outperformed the analogous frequentist estimator in finite samples; validated the method on a real observational dataset (NHANES)
- Extending this work to offline reinforcement learning, establishing calibration of a soft-BART-based posterior over an off-policy value functional (In preparation)

Directed Reading Program, Dept. of Mathematics

Jan 2024 - Jan 2025

University of Iowa

- Year-long directed reading with a graduate student mentor covering measure theory and functional analysis

Publications and Preprints

- Schmidt, T. and Wikle, N. Calibrated Bayesian Inference for Stochastic Intervention Effects. *Under review at Bayesian Analysis.*
- Schmidt, T. and Wikle, N. Semiparametric Posterior Corrections for Off-Policy Evaluation. *In preparation.*

Teaching Experience

Teaching Assistant, STAT 1030: Statistics for Business

Spring 2026, Fall 2026

University of Iowa

Teaching Assistant, STAT 5200: Applied Statistics I

Fall 2026

University of Iowa

Honors and Awards

- University of Iowa President's List: Spring 2025, Fall 2025
- University of Iowa Dean's List: Fall 2023, Fall 2024, Spring 2025, Fall 2025

Technical Skills

- Programming: [Python, R, C++]
- Methods: Bayesian nonparametrics, semiparametric theory, causal inference, reinforcement learning, MCMC, off-policy evaluation